MTH 200 **The Precise Definition of a Limit**

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer each and every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Consider the limit $\lim_{x\to 4}(4x-9)=7$.

a) Suppose that someone challenged you to show that the y-values of $f(x)=4x-9$ can get within 4 units from 5. Determine a value of δ that would make this possible.

b) Suppose that someone challenged you to show that the y-values of $f(x)=4x-9$ can get within 2 units from 5. Determine a value of δ that would make this possible.

c) Suppose that someone challenged you to show that the y-values of $f(x)=4x-9$ can get within $\frac{1}{2}$ units from 5. Determine a value of δ that would make this possible.

d) Suppose that someone challenged you to show that the y-values of $f(x)=4x-9$ can get within ε units from 5. Determine a value of δ that would make this possible.

e) Use the epsilon-delta definition of a limit to prove that $\lim_{x \to 4}(4x-9)=7$.

2. Determine the limit of $\lim_{x \to -1}\left(\frac{5-x}{2}\right)$ and prove the limit is correct.